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Sem: 7

R.No: 15

Div: A

**Sub: Web Development using fullstack
(701)**

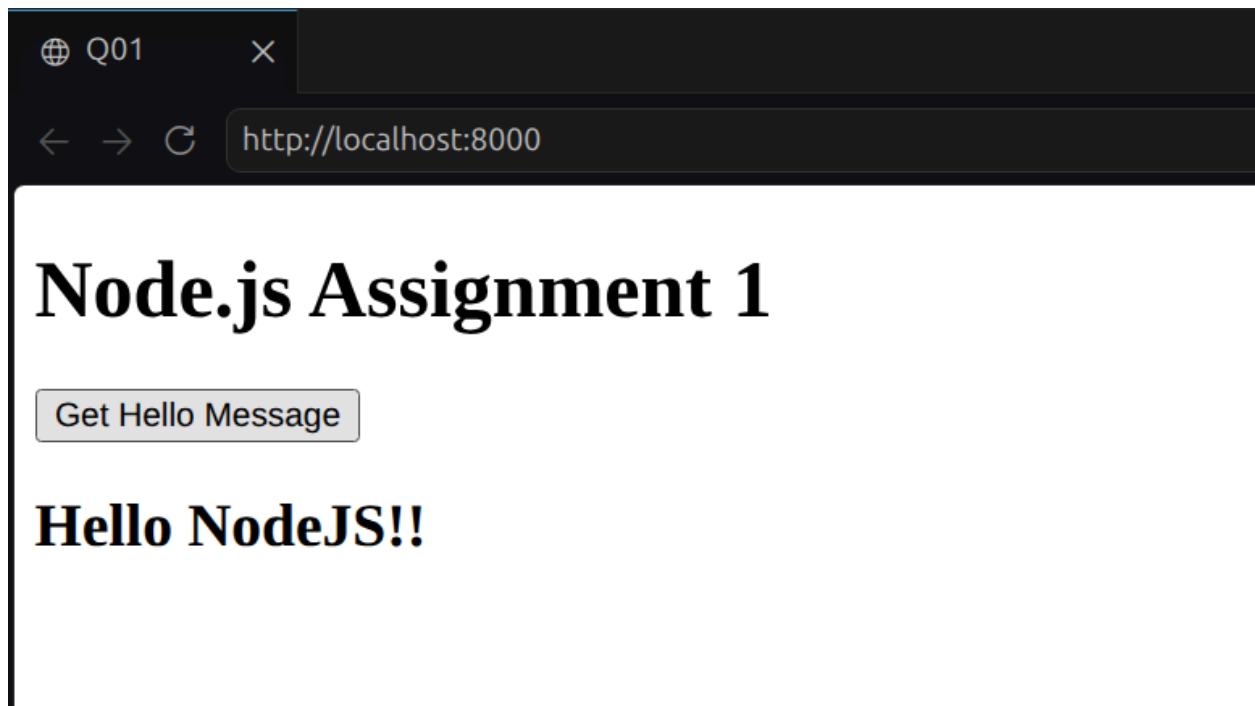
Practical Assignment 1

Question 1

Develop Node.js application with following requirements:

- Develop a route **"/gethello"** with GET method. It displays **"Hello NodeJS!!"** as response.
- Make an HTML page and display.
- Call **"/gethello"** route from HTML page using AJAX call. (Any frontend AJAX API can be used.)

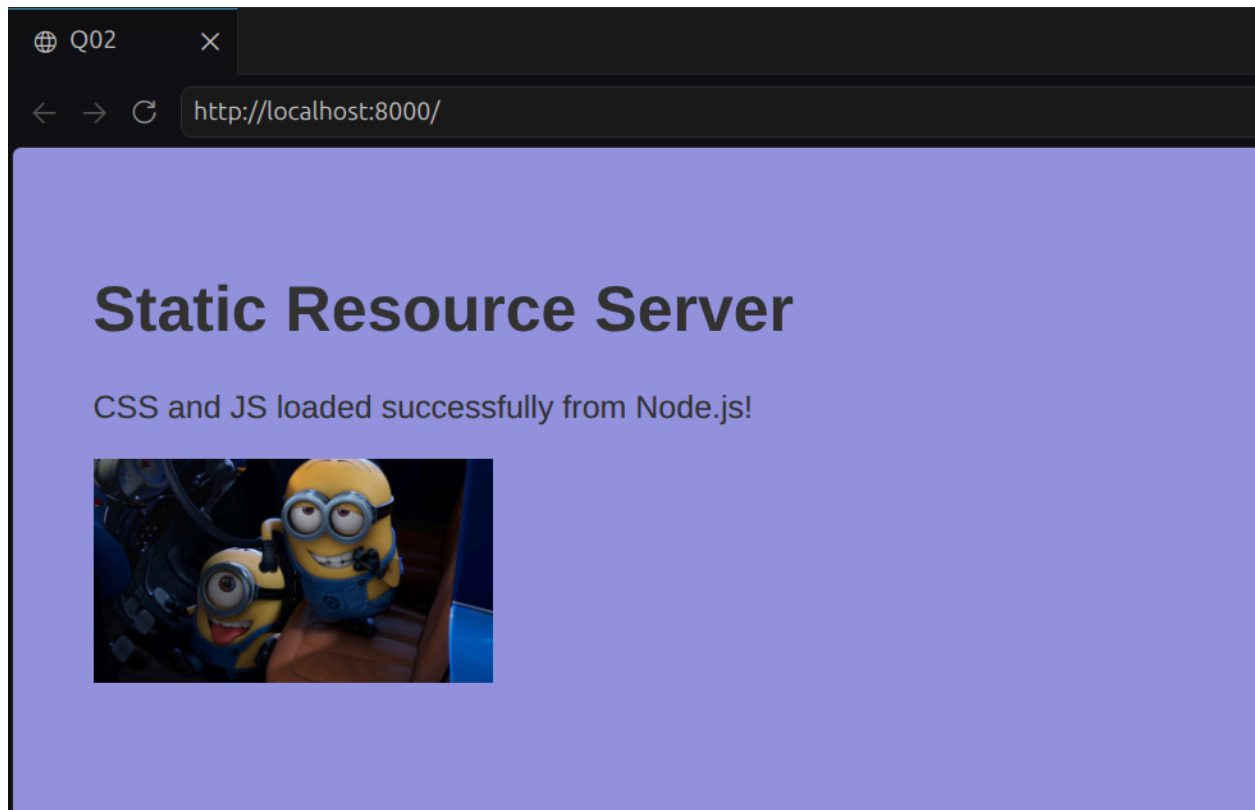
Output Screenshot(s)



Question 2

Develop a web server which serves static resources.

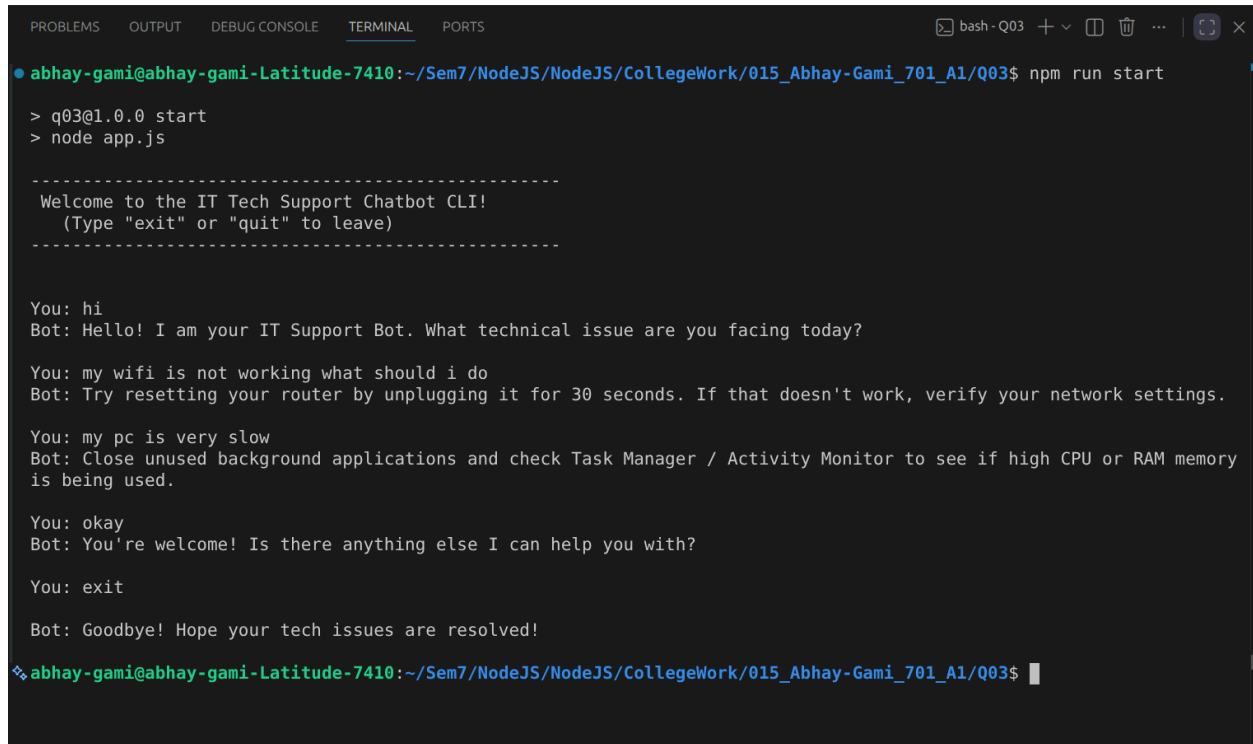
Output Screenshot(s)



Question 3

Develop a module for domain specific chatbot and use it in a command line application.

Output Screenshot(s)



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
bash - Q03 + - [] ... | [X] X

• abhay-gami@abhay-gami-Latitude-7410:~/Sem7/NodeJS/NodeJS/CollegeWork/015_Abhay-Gami_701_A1/Q03$ npm run start

> q03@1.0.0 start
> node app.js

-----
Welcome to the IT Tech Support Chatbot CLI!
(Type "exit" or "quit" to leave)
-----

You: hi
Bot: Hello! I am your IT Support Bot. What technical issue are you facing today?

You: my wifi is not working what should i do
Bot: Try resetting your router by unplugging it for 30 seconds. If that doesn't work, verify your network settings.

You: my pc is very slow
Bot: Close unused background applications and check Task Manager / Activity Monitor to see if high CPU or RAM memory is being used.

You: okay
Bot: You're welcome! Is there anything else I can help you with?

You: exit

Bot: Goodbye! Hope your tech issues are resolved!

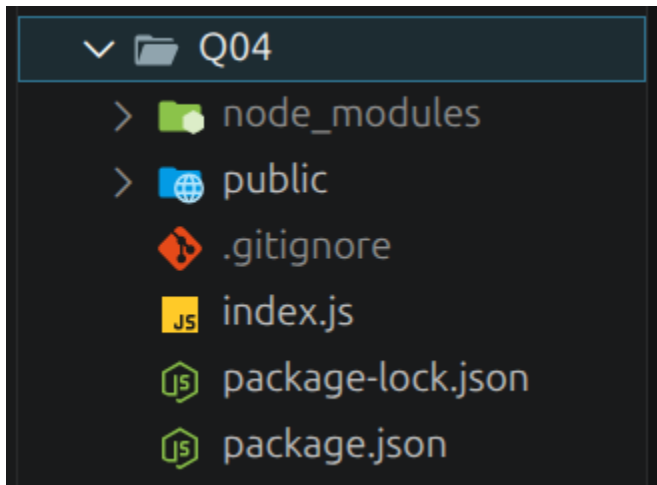
❖ abhay-gami@abhay-gami-Latitude-7410:~/Sem7/NodeJS/NodeJS/CollegeWork/015_Abhay-Gami_701_A1/Q03$
```

Question 4

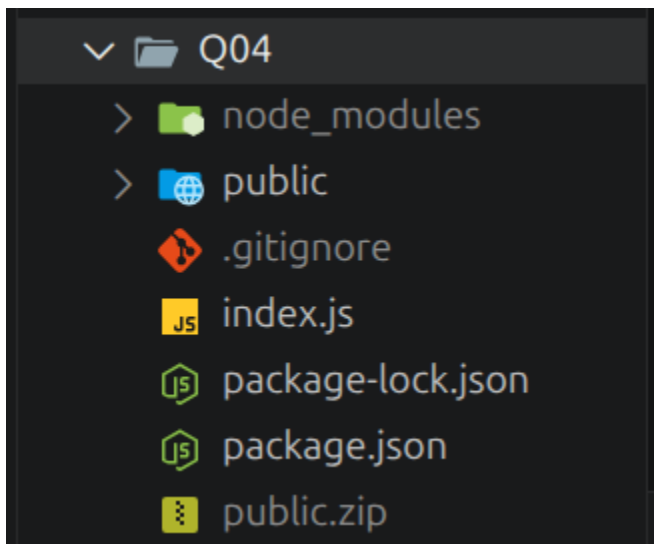
Write a program to create a compressed ZIP file for a folder.

Output Screenshot(s)

Before Running:



After Running:

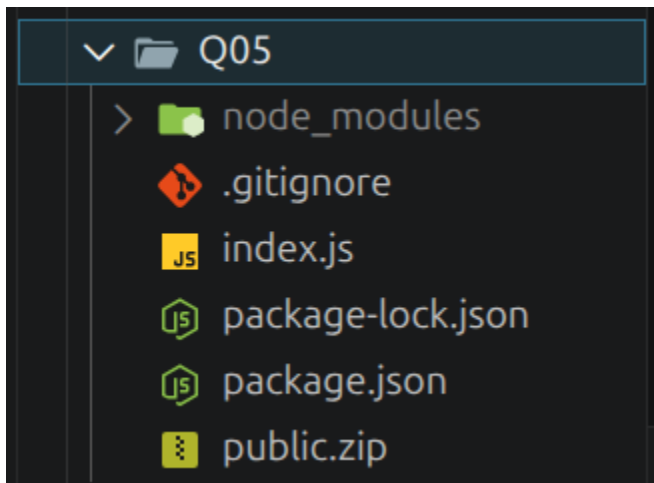


Question 5

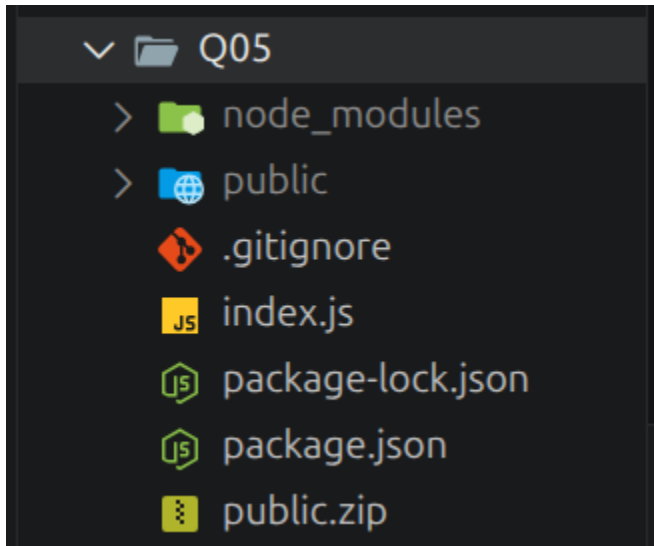
Write a program to extract a ZIP file.

Output Screenshot(s)

Before Running



After Running



Question 6

Write a program to promisify `fs.unlink()` function and call it.

Output Screenshot(s)

```
● abhay-gami@abhay-gami-Latitude-7410:~/Sem7/NodeJS/NodeJS/CollegeWork/015_Abhay-Gami_701_A1/Q06$ npm run start
> q06@1.0.0 start
> node index.js

=== Promisifying fs.unlink Demonstration ===

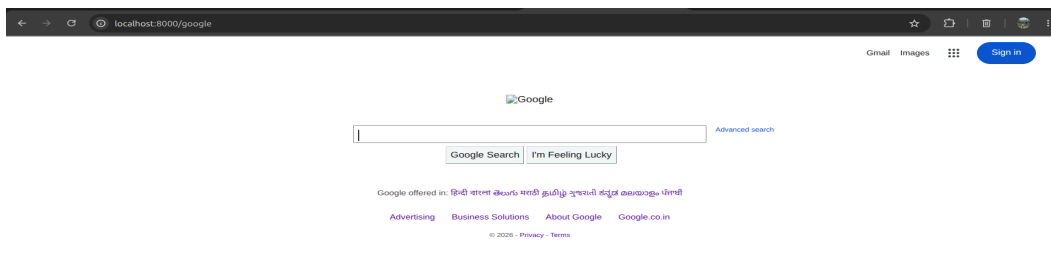
Created temporary file: ./temp_to_delete.txt
Checking if file exists: true
Calling customUnlink()...
Successfully deleted file using Custom Promisified fs.unlink!
Checking if file exists: false

Created temporary file again: ./temp_to_delete.txt
Checking if file exists: true
Calling promisifiedUnlink() from util.promisify...
Successfully deleted file using util.promisify!
Checking if file exists: false
❖ abhay-gami@abhay-gami-Latitude-7410:~/Sem7/NodeJS/NodeJS/CollegeWork/015_Abhay-Gami_701_A1/Q06$
```

Question 7

Develop a Node.js web server with a route `/google` which fetches data from the Google page using `fetch()` with the `async-await` model and displays it as a response.

Output Screenshot(s)



Question 8

Set the following scripts in package.json:

- Server Script
- Test Script
- Three User Defined Scripts

Output Screenshot(s)

```
● abhay-gami@abhay-gami-Latitude-7410:~/Sem7/NodeJS/NodeJS/CollegeWork/015_Abhay-Gami_701_A1/Q08$ npm run test
> q08@1.0.0 test
> node test.js

Running Q08 tests...
✓ Test 1: Port is configured correctly
✓ Test 2: Predefined and user-defined scripts are registered in package.json
All Q08 tests passed successfully!
● abhay-gami@abhay-gami-Latitude-7410:~/Sem7/NodeJS/NodeJS/CollegeWork/015_Abhay-Gami_701_A1/Q08$ npm run server
> q08@1.0.0 server
> node server.js

[Q08] Server running at http://localhost:8000
[Q08] Gracefully shutting down the server...
● abhay-gami@abhay-gami-Latitude-7410:~/Sem7/NodeJS/NodeJS/CollegeWork/015_Abhay-Gami_701_A1/Q08$ npm run info
> q08@1.0.0 info
> echo 'Project Info: Q08 NodeJS custom scripts demonstration'

Project Info: Q08 NodeJS custom scripts demonstration
● abhay-gami@abhay-gami-Latitude-7410:~/Sem7/NodeJS/NodeJS/CollegeWork/015_Abhay-Gami_701_A1/Q08$ npm run greet
> q08@1.0.0 greet
> echo 'Hello from your custom NodeJS script!'

Hello from your custom NodeJS script!
● abhay-gami@abhay-gami-Latitude-7410:~/Sem7/NodeJS/NodeJS/CollegeWork/015_Abhay-Gami_701_A1/Q08$ npm run dev
> q08@1.0.0 dev
> node --watch server.js

[Q08] Server running at http://localhost:8000
[Q08] Gracefully shutting down the server...
Completed running 'server.js'. Waiting for file changes before restarting...
^C
❖ abhay-gami@abhay-gami-Latitude-7410:~/Sem7/NodeJS/NodeJS/CollegeWork/015_Abhay-Gami_701_A1/Q08$
```

Question 9

Write a program which calls useful functions of the fs module.

Output Screenshot(s)

```
abhay-gami@abhay-gami-Latitude-7410:~/Sem7/NodeJS/NodeJS/CollegeWork/015_Abhay-Gami_701_A1/Q09$ npm run start
> q09@1.0.0 start
> node index.js

=== Node.js fs Module Functions Demonstration ===

1. Calling fs.mkdirSync()...
✓ Created directory: ./demo_dir

2. Calling fs.existsSync()...
Is directory present? true

3. Calling fs.writeFileSync()...
✓ Written data to file: demo_dir/example.txt

4. Calling fs.readFileSync()...
✓ File Content:
""
Hello, this is the initial file content.
""

5. Calling fs.appendFileSync()...
✓ Content appended successfully.
✓ Updated File Content:
""
Hello, this is the initial file content.
Adding this line to the file.
""

6. Calling fs.statSync()...
✓ File Stats for demo_dir/example.txt:
- Size: 71 bytes
- Is File: true
- Is Directory: false
- Last Modified: Wed Jul 15 2026 17:51:23 GMT+0530 (India Standard Time)

7. Calling fs.readdirSync()...
✓ Directory './demo_dir' files list: [ 'example.txt' ]

8. Calling fs.renameSync()...
✓ Renamed demo_dir/example.txt -> demo_dir/renamed_example.txt
Files in directory after renaming: [ 'renamed_example.txt' ]

9. Calling fs.unlinkSync()...
✓ File deleted: demo_dir/renamed_example.txt
Files in directory after deleting file: []

10. Calling fs.rmdirSync()...
✓ Directory deleted: ./demo_dir
Is directory present now? false

=== Demonstration Completed Successfully ===
```

Question 10

Write a program which:

- Uses global objects in Node.js.
- Reads command line arguments.
- Prints the command line arguments.

Output Screenshot(s)

```
● abhay-gami@abhay-gami-Latitude-7410:~/Sem7/NodeJS/NodeJS/CollegeWork/015_Abhay-Gami_701_A1/Q10$ npm run start
> q10@1.0.0 start
> node index.js

=== Node.js Global Objects Demonstration ===

1. Directory and File Path Globals:
  - Current Directory (__dirname): /home/abhay-gami/Sem7/NodeJS/NodeJS/CollegeWork/015_Abhay-Gami_701_A1/Q10
  - Current File Path (__filename): /home/abhay-gami/Sem7/NodeJS/NodeJS/CollegeWork/015_Abhay-Gami_701_A1/Q10/index.js

2. Process Global Object details:
  - Node Version: v22.23.1
  - Platform: linux
  - Process ID (PID): 15143
  - Current Directory: /home/abhay-gami/Sem7/NodeJS/NodeJS/CollegeWork/015_Abhay-Gami_701_A1/Q10

3. Console Global Object:
  - Logging via console.log() is done!

4. Global context properties check:
  - Does globalThis exist? true
  - Does global exist? true

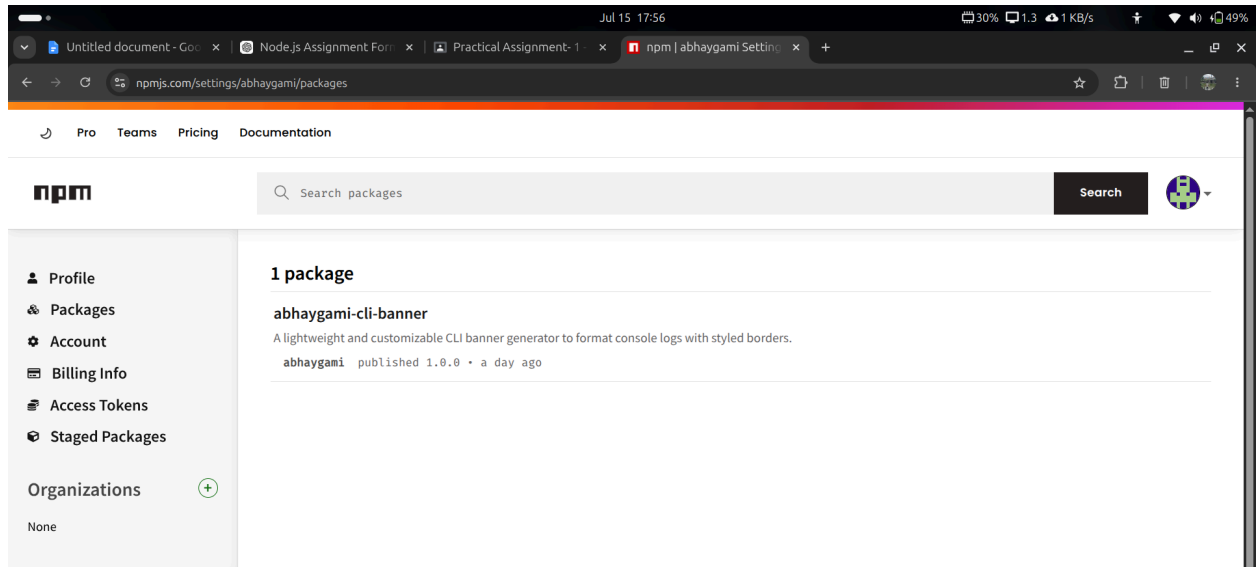
5. Command Line Arguments (process.argv):
  Total argument tokens: 2
  List of arguments:
    [0] /home/abhay-gami/.nvm/versions/node/v22.23.1/bin/node (Node executable path)
    [1] /home/abhay-gami/Sem7/NodeJS/NodeJS/CollegeWork/015_Abhay-Gami_701_A1/Q10/index.js (JavaScript entry script path)

6. Demonstrating setTimeout (Global Async Function)...
✓ Callback inside setTimeout executed after delay! (Global demo finished)
❖ abhay-gami@abhay-gami-Latitude-7410:~/Sem7/NodeJS/NodeJS/CollegeWork/015_Abhay-Gami_701_A1/Q10$
```

Question 11

Develop a useful package and publish it on npmjs.com.
Write a program to use the published package.

Output Screenshot(s)



```
● abhay-gami@abhay-gami-Latitude-7410:~/Sem7/NodeJS/NodeJS/CollegeWork/015_Abhay-Gami_701_A1/Q11/consumer$ npm run start
t
> consumer@1.0.0 start
> node index.js

Starting NPM package consumer demo...

*****
*                                     *
*               WELCOME               *
* -----*
* This banner is generated using the custom abhaygami-cli-banner package! *
* -----*
*****

#####
#          SUCCESS ALERT          #
# -----#
# Operation completed successfully. #
#####

*****
* Simple border message with default formatting. *
*****

● abhay-gami@abhay-gami-Latitude-7410:~/Sem7/NodeJS/NodeJS/CollegeWork/015_Abhay-Gami_701_A1/Q11/consumer$
```

Question 12

Write a Node.js program having at least three dependent tasks and execute them using chained promises.

Output Screenshot(s)

```
● abhay-gami@abhay-gami-Latitude-7410:~/Sem7/NodeJS/NodeJS/CollegeWork/015_Abhay-Gami_701_A1/Q12$ npm run start
> ql2@1.0.0 start
> node index.js

=== Chained Promises Dependent Tasks Demonstration ===

Initiating promise chain with User ID: 101...
[Task 1] Fetching profile for user ID: 101...
✓ [Task 1 Success] Profile details loaded.
[Task 2] Profile retrieved for Abhay Gami. Fetching orders...
✓ [Task 2 Success] Orders successfully fetched.
[Task 3] Generating invoice for 3 items...
✓ [Task 3 Success] Invoice generated successfully.

===== INVOICE DETAILS =====
Invoice No:   INV-319590
Customer:    Abhay Gami (abhay@example.com)
Items:
  - Mechanical Keyboard ($90)
  - Ergonomic Mouse ($50)
  - USB-C Hub ($30)
Total:       $170
=====

Promise chaining execution complete!
❖ abhay-gami@abhay-gami-Latitude-7410:~/Sem7/NodeJS/NodeJS/CollegeWork/015_Abhay-Gami_701_A1/Q12$
```
